



Auto Topup Module Audit

Gnosis Ltd - Report by Côme du Crest

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Auto Topup Module Audit

This document presents the findings of a smart contract audit conducted by Côme du Crest for Gnosis Ltd.

Scope

The scope includes selected contracts within `gnosis/wallet-contracts/autotupup` as of commit `0xf64849f`.

The contracts have been reviewed after commit `0xbe7088c` which updates the authorized caller of `triggerTopup()` from the roles modifier of the app safe to the safe itself.

Context

The goal of the auto top up module is to enable automatic token transfers from a Gnosis App Safe to a Gnosis Pay card safe to be used by the card.

It is assumed that the app safe has a Zodiac roles modifier enabled that allows calling `AutoTopupModule.triggerTopup()`. The `AutoTopupModule` is enabled as a module on the app safe. The function `triggerTopup()` ensures it is called by the app safe Zodiac roles modifier before transferring tokens from the app safe to the card safe via `execTransactionFromModule()`.

The flow is described as follows by the developer:

- Off-chain bot detect balance deficit.
- The bot calls `execTransactionWithRole()` on the enabled Zodiac roles modifier.
- The roles modifier checks that the relayer is authorized, the call is made to `triggerTopup()`, and the daily limit is not reached.
- The Modifier calls the Module.
- The Module checks the caller is the roles modifier, and the balance constraints are respected.
- The Module triggers the `execTransactionFromModule()` on the App Safe to refill the card safe balance.

Status

No issues have been found.

The report has been sent to the core developer.